

1 SCENARIOS — BOWL SEATS + SUITES ≈ 2,200 SPECTATORS (BASE INPUTS)									
FLOOR SF	BOWL	SUITES	SPECT.	SUITE SF	FRONT FT	FOOTPRINT	TOTAL GSF	LVL	FIT
22,000 (locked floor): suites up to 2,200 — NO FIT at 12, 16 or 20 guests, suites on L2 or L3.									
Most spectators that fit (suites on L3, bowl reduced to make room):									
22,000	1,180	38 x 12	1,636	12,933	418	54,992	96,826	3	YES
22,000	1,190	28 x 16	1,638	12,637	392	54,967	96,327	3	YES
22,000	1,180	23 x 20	1,640	12,934	391	54,986	96,812	3	YES
20,000	1,610	50 x 12	2,210	17,017	550	54,987	105,812	3	YES
20,000	1,610	37 x 16	2,202	16,699	518	54,962	105,311	3	YES
20,000	1,610	30 x 20	2,210	16,870	510	54,971	105,566	3	YES
18,000	1,940	22 x 12	2,204	7,487	242	54,994	92,682	3	YES
18,000	1,940	17 x 16	2,212	7,673	238	54,994	92,929	3	YES
18,000	1,940	13 x 20	2,200	7,310	221	54,994	92,313	3	YES
16,400	2,200	0 needed	2,200	0	—	54,949	80,930	2	YES
22,000 lean	2,200	0 needed	2,200	0	—	51,631	74,678	2	YES

Suites on L3 (a suite level just above the top row of the upper tier) unless noted. Suites at the back of the upper tier on Level 2 never fit on base inputs: Level 2 is already as big as the L1 ring under it. Bowl = largest bowl that fits (10-seat steps, best tier split); suites = (target – bowl) ÷ guests, rounded up. SUITE SF = suites + 44 in corridor (net). FRONT FT = total suite frontage. Bold row = headline. Lean inputs fit 2,200 bowl seats with no suites (Rev B).

2 WHY THE 22,000 SF FLOOR CAN'T REACH 2,200 WITH SUITES

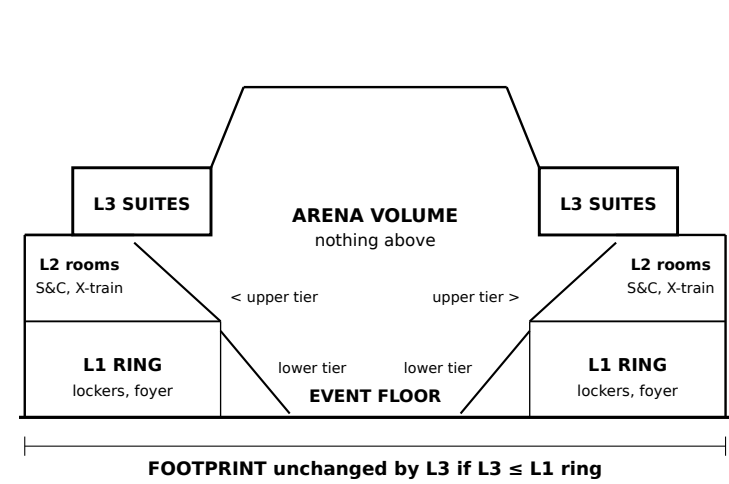
On base inputs the locked floor already fills the 55,000 footprint at ≈ 1,370 bowl seats (Rev B). A bowl seat costs 6.0 SF; a suite guest costs 25 SF plus corridor. Suites on L3 add no footprint by themselves, but the added floor area raises mechanical (5% of all gross, at grade) and stair widths on L1, so bowl seats must come out to make room. Best mix: 1,180 bowl + 23 x 20 = 1,640 spectators. Rooftop mechanical would help; not credited.

3 SUITE MODULE (CITED SIZING, R-016)

GUESTS	SUITE SF	WIDTH FT	CORRIDOR SF	CODE LOAD	WHEELCHAIR	GUEST SOURCE
12	300	11	40	28 1 + companion	Pitt Petersen (12 guests)	
16	400	14	51	37 1 + companion	WKU Diddle (16 tickets)	
20	500	17	62	46 1 + companion	Sheldon ISD (20 VIPs)	

Suite SF = 25 SF/guest: Diddle Arena (WKU) suites 400 SF with 16 tickets; Texas Tech's standard suites are 276 SF (cross-check). Includes seating ledge + lounge. Kitchenette/pantry: no source, left out (served from the concession stand). Width rule (2 ledge rows x 18 in + 2 ft) is ASSUMED. Code load = seats (IBC 1004.6) + lounge ÷ 15 net (T1004.5 unconcentrated): higher than the guest count, used for exits and fixtures. Revenue is out of scope: no \$ figures.

4 WHERE SUITES GO — SECTION (NTS)



L3 suites sit over the L2 rooms and the L1 ring, so they add floor area but not footprint (headline: L3 ≈ 11,300 GSF vs L1 ring ≈ 26,000). The catch: every added floor still adds mechanical (5% of gross, at grade) and wider stairs on L1, and suites need 25 SF per guest vs 6 SF per bowl seat. On Level 2 (back of the upper tier) suites never fit: L2 is already ≈ the size of the L1 ring.

5 A SUITE LEVEL = A 3RD STORY

- Headline case: L2 load 1,368, L3 (suites, code load) 598 -> 4 stairs x 68 in on all 3 levels (T1006.3.3; 0.2 in/occ., 1005.3.1; width never reduced downstream, 1005.4).
- Open stairs lose the two-story exception (1019.3 exc. 1): enclosed exit stairs, 1-hour for fewer than 4 stories (1023.2).
 - Elevator must stop at the suite level (1104.4); a wheelchair space + companion seat in every suite (1109.2.2.2, ADA 221.2.1.2), with lines of sight and dispersion (ADA 221.2.3, IBC 1109.2.4).
 - Elevator egress still not required (< 4 stories, 1009.2.1). Allowed stories for the construction type (IBC ch. 5) NOT checked.
 - Could the suites be a mezzanine instead of a story? Not checked (architect).

6 PRECEDENTS (R-016)

- Diddle Arena (WKU): 16 suites, 400 SF, 16 tickets each, restrooms for the suite level, suites 'clear at the top of the arena' (2002).
- United Supermarkets Arena (Texas Tech): 24 suites at 276 SF + corner suites at 611 SF.
- Petersen Events Center (Pitt): 5 courtside suites up to 15; 16 smaller suites, about 12 guests.
- Sheldon ISD Panther Stadium (high school): 2 suites, 20 VIPs each; lounge seating, sink, refrigerator; restrooms on the level.

7 RESULT (BASE INPUTS)

18,000 SF FLOOR + SUITES:

1,940 bowl + 13 suites x 20 = 2,200

Footprint ≈ 54,994 vs 55,000: FITS

Total ≈ 92,300 GSF · 3 levels (suite level)

At 22,000 SF: suites reach only ≈ 1,640

Largest floor that still reaches 2,200 with suites: ≈ 20,200-20,300 SF. At 20,000 SF it takes 30-50 suites (≈ 510-550 ft of suite front), beyond the cited arenas' 16-24 suites. 18,000 SF needs 13-22 suites (≈ 220-240 ft, both long sides). The headline is ≈ 6 SF under the cap: no margin.

8 RESTROOMS + WHEELCHAIR SPACES

T2902.1 per level (headline): L1 30, L2 30, L3 suites 16 fixtures (WC 4 M / 8 W, lav 2 / 2) = 76 total (Rev B: 70). Suite fixtures use the suite code load, not the guest count. Wheelchair spaces: bowl 16 (T1109.2.2.1) + 1 in each of 13 suites.

9 CITED vs ASSUMED

CITED: 12/16/20 guests, 400 SF suite (25 SF/guest), suite-level restrooms, suites at the top of the bowl (Diddle), IBC 1004.6, T1004.5, T1020.3, 1019.3, 1023.2, 1104.4, 1109.2, ADA 221.2. ASSUMED: suite width rule, 44 in corridor, suite level over the ring, no kitchenette, mechanical at grade, all Rev B assumptions.

10 STILL OPEN (D-030)

Pick one: (a) 18,000 SF floor + 13-22 suites, 3 levels; (b) 16,400 SF floor, all 2,200 in the bowl, no suites, 2 levels (≈ 80,900 GSF); (c) keep 22,000 SF with lean inputs (telescopic lower tier), no suites, 2 levels (≈ 74,700 GSF). ~1,370 seats alone is rejected (D-032).

11 SOURCES (retrieved 2026-10-03)

- WKU College Heights Herald 2002-10-22 (Diddle Arena suites); Texas Tech United Supermarkets Arena facts; Petersen Events Center Production Guide 2020; Sheldon ISD Panther Stadium info — R-016
- IBC 2021 (UpCodes) ch. 10, 11; 2010 ADA Standards 221.2 — R-016, R-015
- Rev A/B sources: R-008, R-009, R-014, R-015

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	PROGRAM TEST-FIT, SUITES FOOTPRINT MATH, NOT A PLAN	NTS	October 2026	C	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-G-003